RELATIVISTIC ENERGY / TAYLOR SERIES

ONE ENERGY. MANY TERMS.

REST ENERGY

$$E_0 = Mc^2$$

The baseline

TOTAL ENERGY

$$E = \gamma Mc^2$$

Including motion

Newtonian motion emerges.

One expansion, term by term.

$$E = \frac{Mc^2}{\sqrt{1 - v^2/c^2}}$$

Fixed rest mass M > 0. Speed measured in one inertial frame.

START EXACT.
CHOOSE THE VARIABLE.

Expand about zero speed.

THE EXACT STARTING POINT

$$E = \gamma M c^2, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

REST ENERGY

$$E_0 = Mc^2$$

SPEED RATIO

$$\beta = \frac{v}{c}$$

VARIABLE

$$x = \beta^2$$

$$\frac{E}{E_0} = f(x) = (1 - x)^{-1/2}$$

M is invariant mass; c is the speed of light.

E and v belong to the same inertial frame.

Small x means speed well below c.

DIFFERENTIATE.
REVEAL THE PATTERN.

Each derivative at zero is divided by its factorial.

$$f(x) = (1 - x)^{-1/2}$$

DERIVATIVE

$$f(0) = 1$$

$$f'(0) = \frac{1}{2}$$

$$f''(0) = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$$

$$f'''(0) = \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{2}$$

TAYLOR COEFFICIENT

$$a_0 = 1$$

$$a_1 = \frac{1/2}{1!} = \frac{1}{2}$$

$$a_2 = \frac{3/4}{2!} = \frac{3}{8}$$

$$a_3 = \frac{15/8}{3!} = \frac{5}{16}$$

THE GENERAL COEFFICIENT

$$a_n = \frac{f^{(n)}(0)}{n!} = \frac{(2n)!}{4^n (n!)^2}$$

SUBSTITUTE.
RESTORE THE UNITS.

Use $x = v^2/c^2$, then multiply every term by Mc^2.

01 EXPAND THE DIMENSIONLESS FUNCTION

$$(1-x)^{-1/2} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \cdots$$

02 PUT THE SPEED BACK IN

$$E = Mc^2\left[1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4} + \frac{5}{16}\frac{v^6}{c^6} + \cdots\right]$$

03 THE ENERGY SERIES

$$E = Mc^2 + \frac{1}{2}Mv^2 + \frac{3}{8}\frac{Mv^4}{c^2} + \frac{5}{16}\frac{Mv^6}{c^4} + \cdots$$

Every term has units of energy.

Even powers appear because $x = (v/c)^2$.

READ THE TERMS. REST AND MOTION.

Total energy separates into a baseline and kinetic energy.

REST

PRESENT AT ZERO SPEED

$$E_0 = Mc^2$$

Fixed by the invariant mass.
It remains when v = 0.

NORMALIZED BASELINE

$$\frac{E_0}{Mc^2} = 1$$

REST ENERGY

$$E(v=0) = Mc^2$$

MOTION

KINETIC ENERGY

$$K \equiv E - E_0$$

The energy added by motion.
It vanishes when v = 0.

EXACT KINETIC ENERGY

$$K \equiv (\gamma - 1)Mc^2$$

LOW-SPEED LIMIT

$$K \approx \frac{1}{2}Mv^2$$

Higher powers of v/c supply the relativistic corrections.
These are expansion terms, not quantized energy levels.

THE WHOLE SERIES. THE PRACTICAL LIMIT.

An infinite sum and a finite approximation have different roles.

INFINITE SERIES

$$\frac{E}{E_0} = \sum_{n=0}^{\infty} a_n \beta^{2n}$$

Exact for |v| < c.

FINITE TRUNCATION

$$\frac{E}{E_0} \approx 1 + \tfrac{1}{2}\beta^2 + \tfrac{3}{8}\beta^4$$

Most useful when v ≪ c.

HOW MUCH DOES A SHORT SUM MISS?

Speed v/c	Exact E/E₀	Through v⁴	Error*
0.1	1.0050378	1.0050375	0.00003%
0.5	1.1547005	1.1484375	0.54%
0.9	2.2941573	1.6510375	28.03%

*Percent below the exact total energy; values rounded.

All omitted terms are positive for 0 < v < c.

As v approaches c, more terms are needed.

KEEP THE ENERGY.
KEEP THE DISTINCTIONS.

01 **Total energy includes rest energy.**

Subtract Mc^2 to isolate kinetic energy.

02 **Newtonian motion is the leading term.**

$K \approx \frac{1}{2}Mv^2$ when v is much smaller than c.

03 **The full series has a domain.**

The series converges for $|v| < c$; short sums approximate.

REFERENCES / KEEP EXPLORING

[1] NIST DLMF, §4.6(ii), Eq. 4.6.7
Binomial expansion and its convergence domain.
dlmf.nist.gov/4.6#E7

[2] Einstein (1905), On the Electrodynamics of Moving Bodies, §10.
Kinetic energy; see also the September mass-energy paper.
English translations: fourmilab.ch/etexts/einstein/

Modern notation and derivation. Free massive particle; no gravity.

Save the equations. Keep asking why.